



Final Theoretical Examination
Year: 2025-2026 Term: First
Subject: Biostatistics
Code: M 255
Students: 2nd year students
Program: Biochemistry

Mathematics of Department
Time: Two hour
Staff: Dr. M. Elmenshawy
Date: 22/1/2025
No. of Questions: 4
No. of Pages: 1 Total Marks: 80



INSTRUCTIONS: The use of mobile phones is not allowed, use blue or black ink only, no departures until at least half the allocated time has passed, Show your ID upon request, course coordinator can use this part for other instructions as calculator use.

Answer the following questions:

Question 1:

(20 marks)

- a) Acetone levels were normally distributed with mean of 870 and standard deviation 211 ppb. Find the probability that on a given day the subject's acetone level is:
1. Between 600 and 1000 ppb
 2. Over 900 ppb
 3. Under 500 ppb
- b) Assume that the weights of all packages of a certain brand of cookies are normally distributed with a mean of 32 ounces and a standard deviation of .3 ounce. Find the probability that the mean weight, of a random sample of 20 packages of this brand of cookies will be between 31.8 and 31.9 ounce.

Question 2:

(20 marks)

- a) The average zinc concentration recorded from a sample of zinc measurements in 36 different locations is found to be 2.6 gram/milliliter. Assume that the population standard deviation is 0.3,
1. Find the point estimate of population mean?
 2. Find 95% confidence interval for the mean zinc concentration in the river?
- b) The following information was obtained from two independent samples selected from two normally distributed populations with unknown but equal standard deviation

$$n_1 = 28, \bar{x}_1 = 47, S_1 = 8$$
$$n_2 = 29, \bar{x}_2 = 45, S_2 = 6$$

1. Construct a 98% confidence interval for $\mu_1 - \mu_2$?
2. What is the margin error of estimate $\mu_1 - \mu_2$?

Question 3

(20 marks)

- a- A builder claims that heat pumps are installed in 70% of all homes being constructed today in the city of Cairo. Would you agree with this claim if a random survey of new homes in the city shows that 8 out of 15 homes had heat pumps installed? Use a 0.1 level of significance.
- b- A sample of 40 children from New York State showed that the mean time they spend watching television is 28.50 hours per week with a standard deviation of 4 hours. Another sample of 35 children from California showed that the mean time spent by them watching television is 23.25 hours per week with a standard deviation of 5 hours. Using a 2.5% significance level, can you conclude that the mean time spent watching television by children in New York State is greater than that for children in California? Assume that the standard deviations for the two populations are equal.

Question 4**(20 marks)**

The Table below show that the life time under controlled condition, in hours in excess of 1000 hours, of samples of 60w of electric light bulbs of three different brands

Brand I	Brand II	Brand III
16	18	26
15	22	31
13	20	24
21	16	30
15	24	24

Assuming all life time to be normally distributed with common variance, test at the 1% significance level hypothesis that there is no difference between the three brands with respect to mean lifetime.

Hint 1: Standard normal distribution table

z-value	0.02	0.04	0.05	0.06	0.08	0.09
-2.9					0.0014	
-1.7			0.0401			
-1.4						0.0681
-1.2					0.0985	
0.1		0.5557				
0.6	0.7324					
1.6			0.950			
1.9				0.975		

Hint 2: t-distribution table

Df	0.025	0.01
55		2.396
73	1.993	

Hint 3: Some values from F-distribution table

$$F_{0.01,(2,12)} = 6.92, F_{0.01,(12,2)} = 99.42, F_{0.05,(2,12)} = 3.89, F_{0.05,(12,2)} = 19.42$$

Course Coordinator

With best wishes

Head of department

Signature

Signature

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Answer the following questions:

Question 1:

(20 marks)

- Define the sampling distribution.
 - An electric firm manufactures light bulbs that have a length of life that is approximately normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours.
 - The contents of 7 similar containers of sulfuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, and 9.6 liters. Find a 95% C.I. for the mean of all such containers, assuming an approximate normal distribution.
- Hint: $P(Z < -2.5) = 0.0062$ and $P(Z < 2.5) = 0.9938$, $t_{0.025,6} = 2.447$.

Question 2:

(20 marks)

- A food company is planning to market a new kind of cereal. However, before marketing this product, the company wants to find what percentage of people will like it. The company's research department selected a random sample of 500 persons and asked them to taste this cereal. Of these persons, 290 said they liked it.
 - What is the point estimate of population proportion? What is the margin error of this estimate?
 - Find, with a 99% C.I. for percentage of all people will like this cereal?
- The following information was obtained from two independent samples selected from two normally distributed populations with unknown but equal standard deviation

$$\begin{aligned} n_1 &= 28, & \bar{X}_1 &= 47, & S_1 &= 8 \\ n_2 &= 29, & \bar{X}_2 &= 45, & S_2 &= 6 \end{aligned}$$

Construct a 98% confidence interval for $\mu_1 - \mu_2$? Hint: $Z_{0.025} = 1.96$, $Z_{0.005} = 2.58$, $t_{0.01,55} = 2.396$.

Question 3

(20 marks)

If a random sample of 12 homes included in a planned study of indicates that vacuum cleaners expend an average of 42 kilowatt-hours per year with a standard deviation of 11.9 kilowatt-hours, does this suggest at the 0.05 level of significance that the vacuum cleaners expend, on the average, less than 46 kilowatt-hours annually? Assume the population of kilowatt-hours to be normal?

Hint: $t_{0.05,11} = -1.796$.

Question 4

(20 marks)

The Table below show that the life time under controlled condition, in hours in excess of 1000 hours, of samples of 60w of electric light bulbs of three different brands

Brand I	Brand II	Brand III
16	18	26
15	22	31
13	20	24
21	16	30
15	24	24

Assuming all life time to be normally distributed with common variance, test at the 1% significance level hypothesis that there is no difference between the three brands with respect to mean lifetime.

Hint: $F_{0.01,(2,12)} = 6.92$, $F_{0.01,(12,2)} = 99.42$, $F_{0.05,(2,12)} = 3.89$,
 $F_{0.05,(12,2)} = 19.42$

Course Coordinator

Signature

With best wishes

Head of department

Signature



الإجابات الكاملة المنظمة الدقيقة ستحظى بالدرجات النهائية

اجب عن الأسئلة الآتية

السؤال الأول (20 درجة)

- أخذت عينة عشوائية حجمها 10 من مجتمع طبيعية تباينه σ_1^2 فكان الانحراف المعياري المحسوب من العينة 0.5 وأخذت عينة عشوائية أخرى حجمها 6 من مجتمع طبيعي آخر تباينه σ_2^2 فكان الانحراف المعياري من العينة 0.4
- 1- اختبر الفرض $H_1: \sigma_1^2 > \sigma_2^2$ حيث $(1 - \alpha = 0.95)$.
- 2- أوجد 99% فترة ثقة لتباين المجتمع الثانى σ_2^2 .

السؤال الثانى (20 درجة)

- 1- البيانات الآتية تمثل نتيجة فحص أجهزة من ثلاث مؤسسات مختلفة إلى جهة ما

المؤسسات	نوعية الأجهزة			المجموع
	مرفوضة	مقبولة	ممتازة	
المؤسسة الأولى	12	23	89	
المؤسسة الأولى	8	12	62	
المؤسسة الأولى	21	30	119	
المجموع				

اختبر ما إذا كان هناك استقلال بين نوعية الأجهزة و المؤسسات بمستوى معنوية 0.05 .

2- من مجتمع مكون من 10000 شخص. احسب حجم العينة اللازمة لتقدير نسبة عدد المصابين بمرض معين في ذلك المجتمع بخطأ لا يتعدى 0.03 وبدرجة ثقة 95%.

السؤال الثالث (20 درجة)

- 1- فى دراسة لقياس مستوى انتشار لعقار معين فى الدم صممت تجربته لقياس هذا المستوى عند لحظة الحقن X و بعد 30 دقيقة من الحقن Y لعدد 11 من حيوانات التجارب فكانت النتائج التالية مقاسه بـ (ng/ml × 100)

X	256	518	305	410	705	660	479	739	730	1178	390
Y	702	310	544	399	521	1026	1391	1853	791	485	1110

بفرض ان تلك البيانات من مجتمعات طبيعيه اوجد 90% فترة ثقة للفرق بين متوسطى المجتمعين عند لحظة الحقن و بعد 30 دقيقة من الحقن

- 2- سحبت عينة عشوائية حجمها 800 نبات من أفراد مجتمع مكون من 30000 نبات . فوجد أن عدد المصابين في العينة بمرض معين 480 نبات.

- احسب 95% فترة ثقة لنسبة المصابين بذلك المرض في ذلك المجتمع.

- احسب 95% فترة ثقة لعدد الأفراد المصابين بذلك المرض في ذلك المجتمع.

السؤال الرابع (20 درجة)

- 1- إذا كان الانحراف المعياري لأطوال مجتمع الطلبة هو 62 و أننا اخترنا عليه مكونة من 9 طلاب فكانت

أطوالهم كما يلي: 163, 163, 165, 168, 169, 173, 174, 176, 179, 183, 185, 189, 193, 195, 197, 199, 201, 203, 205, 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 239, 241, 243, 245, 247, 249, 251, 253, 255, 257, 259, 261, 263, 265, 267, 269, 271, 273, 275, 277, 279, 281, 283, 285, 287, 289, 291, 293, 295, 297, 299, 301, 303, 305, 307, 309, 311, 313, 315, 317, 319, 321, 323, 325, 327, 329, 331, 333, 335, 337, 339, 341, 343, 345, 347, 349, 351, 353, 355, 357, 359, 361, 363, 365, 367, 369, 371, 373, 375, 377, 379, 381, 383, 385, 387, 389, 391, 393, 395, 397, 399, 401, 403, 405, 407, 409, 411, 413, 415, 417, 419, 421, 423, 425, 427, 429, 431, 433, 435, 437, 439, 441, 443, 445, 447, 449, 451, 453, 455, 457, 459, 461, 463, 465, 467, 469, 471, 473, 475, 477, 479, 481, 483, 485, 487, 489, 491, 493, 495, 497, 499, 501, 503, 505, 507, 509, 511, 513, 515, 517, 519, 521, 523, 525, 527, 529, 531, 533, 535, 537, 539, 541, 543, 545, 547, 549, 551, 553, 555, 557, 559, 561, 563, 565, 567, 569, 571, 573, 575, 577, 579, 581, 583, 585, 587, 589, 591, 593, 595, 597, 599, 601, 603, 605, 607, 609, 611, 613, 615, 617, 619, 621, 623, 625, 627, 629, 631, 633, 635, 637, 639, 641, 643, 645, 647, 649, 651, 653, 655, 657, 659, 661, 663, 665, 667, 669, 671, 673, 675, 677, 679, 681, 683, 685, 687, 689, 691, 693, 695, 697, 699, 701, 703, 705, 707, 709, 711, 713, 715, 717, 719, 721, 723, 725, 727, 729, 731, 733, 735, 737, 739, 741, 743, 745, 747, 749, 751, 753, 755, 757, 759, 761, 763, 765, 767, 769, 771, 773, 775, 777, 779, 781, 783, 785, 787, 789, 791, 793, 795, 797, 799, 801, 803, 805, 807, 809, 811, 813, 815, 817, 819, 821, 823, 825, 827, 829, 831, 833, 835, 837, 839, 841, 843, 845, 847, 849, 851, 853, 855, 857, 859, 861, 863, 865, 867, 869, 871, 873, 875, 877, 879, 881, 883, 885, 887, 889, 891, 893, 895, 897, 899, 901, 903, 905, 907, 909, 911, 913, 915, 917, 919, 921, 923, 925, 927, 929, 931, 933, 935, 937, 939, 941, 943, 945, 947, 949, 951, 953, 955, 957, 959, 961, 963, 965, 967, 969, 971, 973, 975, 977, 979, 981, 983, 985, 987, 989, 991, 993, 995, 997, 999, 1001, 1003, 1005, 1007, 1009, 1011, 1013, 1015, 1017, 1019, 1021, 1023, 1025, 1027, 1029, 1031, 1033, 1035, 1037, 1039, 1041, 1043, 1045, 1047, 1049, 1051, 1053, 1055, 1057, 1059, 1061, 1063, 1065, 1067, 1069, 1071, 1073, 1075, 1077, 1079, 1081, 1083, 1085, 1087, 1089, 1091, 1093, 1095, 1097, 1099, 1101, 1103, 1105, 1107, 1109, 1111, 1113, 1115, 1117, 1119, 1121, 1123, 1125, 1127, 1129, 1131, 1133, 1135, 1137, 1139, 1141, 1143, 1145, 1147, 1149, 1151, 1153, 1155, 1157, 1159, 1161, 1163, 1165, 1167, 1169, 1171, 1173, 1175, 1177, 1179, 1181, 1183, 1185, 1187, 1189, 1191, 1193, 1195, 1197, 1199, 1201, 1203, 1205, 1207, 1209, 1211, 1213, 1215, 1217, 1219, 1221, 1223, 1225, 1227, 1229, 1231, 1233, 1235, 1237, 1239, 1241, 1243, 1245, 1247, 1249, 1251, 1253, 1255, 1257, 1259, 1261, 1263, 1265, 1267, 1269, 1271, 1273, 1275, 1277, 1279, 1281, 1283, 1285, 1287, 1289, 1291, 1293, 1295, 1297, 1299, 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2297, 2299, 2301, 2303, 2305, 2307, 2309, 2311, 2313, 2315, 2317, 2319, 2321, 2323, 2325, 2327, 2329, 2331, 2333, 2335, 2337, 2339, 2341, 2343, 2345, 2347, 2349, 2351, 2353, 2355, 2357, 2359, 2361, 2363, 2365, 2367, 2369, 2371, 2373, 2375, 2377, 2379, 2381, 2383, 2385, 2387, 2389, 2391, 2393, 2395, 2397, 2399, 2401, 2403, 2405, 2407, 2409, 2411, 2413, 2415, 2417, 2419, 2421, 2423, 2425, 2427, 2429, 2431, 2433, 2435, 2437, 2439, 2441, 2443, 2445, 2447, 2449, 2451, 2453, 2455, 2457, 2459, 2461, 2463, 2465, 2467, 2469, 2471, 2473, 2475, 2477, 2479, 2481, 2483, 2485, 2487, 2489, 2491, 2493, 2495, 2497, 2499, 2501, 2503, 2505, 2507, 2509, 2511, 2513, 2515, 2517, 2519, 2521, 2523, 2525, 2527, 2529, 2531, 2533, 2535, 2537, 2539, 2541, 2543, 2545, 2547, 2549, 2551, 2553, 2555, 2557, 2559, 2561, 2563, 2565, 2567, 2569, 2571, 2573, 2575, 2577, 2579, 2581, 2583, 2585, 2587, 2589, 2591, 2593, 2595, 2597, 2599, 2601, 2603, 2605, 2607, 2609, 2611, 2613, 2615, 2617, 2619, 2621, 2623, 2625, 2627, 2629, 2631, 2633, 2635, 2637, 2639, 2641, 2643, 2645, 2647, 2649, 2651, 2653, 2655, 2657, 2659, 2661, 2663, 2665, 2667, 2669, 2671, 2673, 2675, 2677, 2679, 2681, 2683, 2685, 2687, 2689, 2691, 2693, 2695, 2697, 2699, 2701, 2703, 2705, 2707, 2709, 2711, 2713, 2715, 2717, 2719, 2721, 2723, 2725, 2727, 2729, 2731, 2733, 2735, 2737, 2739, 2741, 2743, 2745, 2747, 2749, 2751, 2753, 2755, 2757, 2759, 2761, 2763, 2765, 2767, 2769, 2771, 2773, 2775, 2777, 2779, 2781, 2783, 2785, 2787, 2789, 2791, 2793, 2795, 2797, 2799, 2801, 2803, 2805, 2807, 2809, 2811, 2813, 2815, 2817, 2819, 2821, 2823, 2825, 2827, 2829, 2831, 2833, 2835, 2837, 2839, 2841, 2843, 2845, 2847, 2849, 2851, 2853, 2855, 2857, 2859, 2861, 2863, 2865, 2867, 2869, 2871, 2873, 2875, 2877, 2879, 2881, 2883, 2885, 2887, 2889, 2891, 2893, 2895, 2897, 2899, 2901, 2903, 2905, 2907, 2909, 2911, 2913, 2915, 2917, 2919, 2921, 2923, 2925, 2927, 2929, 2931, 2933, 2935, 2937, 2939, 2941, 2943, 2945, 2947, 2949, 2951, 2953, 2955, 2957, 2959, 2961, 2963, 2965, 2967, 2969, 2971, 2973, 2975, 2977, 2979, 2981, 2983, 2985, 2987, 2989, 2991, 2993, 2995, 2997, 2999, 3001, 3003, 3005, 3007, 3009, 3011, 3013, 3015, 3017, 3019, 3021, 3023, 3025, 3027, 3029, 3031, 3033, 3035, 3037, 3039, 3041, 3043, 3045, 3047, 3049, 3051, 3053, 3055, 3057, 3059, 3061, 3063, 3065, 3067, 3069, 3071, 3073, 3075, 3077, 3079, 3081, 3083, 3085, 3087, 3089, 3091, 3093, 3095, 3097, 3099, 3101, 3103, 3105, 3107, 3109, 3111, 3113, 3115, 3117, 3119, 3121, 3123, 3125, 3127, 3129, 3131, 3133, 3135, 3137, 3139, 3141, 3143, 3145, 3147, 3149, 3151, 3153, 3155, 3157, 3159, 3161, 3163, 3165, 3167, 3169, 3171, 3173, 3175, 3177, 3179, 3181, 3183, 3185, 3187, 3189, 3191, 3193, 3195, 3197, 3199, 3201, 3203, 3205, 3207, 3209, 3211, 3213, 3215, 3217, 3219, 3221, 3223, 3225, 3227, 3229, 3231, 3233, 3235, 3237, 3239, 3241, 3243, 3245, 3247, 3249, 3251, 3253, 3255, 3257, 3259, 3261, 3263, 3265, 3267, 3269, 3271, 3273, 3275, 3277, 3279, 3281, 3283, 3285, 3287, 3289, 3291, 3293, 3295, 3297, 3299, 3301, 3303, 3305, 3307, 3309, 3311, 3313, 3315, 3317, 3319, 3321, 3323, 3325, 3327, 3329, 3331, 3333, 3335, 3337, 3339, 3341, 3343, 3345, 3347, 3349, 3351, 3353, 3355, 3357, 3359, 3361, 3363, 3365, 3367, 3369, 3371, 3373, 3375, 3377, 3379, 3381, 3383, 3385, 3387, 3389, 3391, 3393, 3395, 3397, 3399, 3401, 3403, 3405, 3407, 3409, 3411, 3413, 3415, 3417, 3419, 3421, 3423, 3425, 3427, 3429, 3431, 3433, 3435, 3437, 3439, 3441, 3443, 3445, 3447, 3449, 3451, 3453, 3455, 3457, 3459, 3461, 3463, 3465, 3467, 3469, 3471, 3473, 3475, 3477, 3479, 3481, 3483, 3485, 3487, 3489, 3491, 3493, 3495, 3497, 3499, 3501, 3503, 3505, 3507, 3509, 3511, 3513, 3515, 3517, 3519, 3521, 3523, 3525, 3527, 3529, 3531, 3533, 3535, 3537, 3539, 3541, 3543, 3545, 3547, 3549, 3551, 3553, 3555, 3557, 3559, 3561, 3563, 3565, 3567, 3569, 3571, 3573, 3575, 3577, 3579, 3581, 3583, 3585, 3587, 3589, 3591, 3593, 3595, 3597, 3599, 3601, 3603, 3605, 3607, 3609, 3611, 3613, 3615, 3617, 3619, 3621, 3623, 3625, 3627, 3629, 3631, 3633, 3635, 3637, 3639, 3641, 3643, 3645, 3647, 3649, 3651, 3653, 3655, 3657, 3659, 3661, 3663, 3665, 3667, 3669, 3671, 3673, 3675, 3677, 3679, 3681, 3683, 3685, 3687, 3689, 3691, 3693, 3695, 3697, 3699, 3701, 3703, 3705, 3707, 3709, 3711, 3713, 3715, 3717, 3719, 3721, 3723, 3725, 3727, 3729, 3731, 3733, 3735, 3737, 3739, 3741, 3743, 3745, 3747, 3749, 3751, 3753, 3755, 3757, 3759, 3761, 3763, 3765, 3767, 3769, 3771, 3773, 3775, 3777, 3779, 3781, 3783,



2,06
11,233

$(n-1)s^2$
57,2498

الإجابات الكاملة المنظمة الدقيقة ستحظى بالدرجات النهائية

اجب عن الأسئلة الآتية
السؤال الأول (20 درجة)

(أ) أخذت عينة عشوائية حجمها 9 من مجتمع طبيعى تباينه σ^2 وكانت قيم العينة هي 75, 74, 69, 76, 71, 66, 73, 68, 67. اختبر الفرض القائل بأن هذه العينة مسحوبة من مجتمع تباينه 6 ($\alpha = 0.05$).

(ب) لدراسة تواجد نوع معين من الكائنات الدقيقة يتم ذلك مجهريا وتقدير مساحتها (cm²) أخذت عينة بحجم 10 وكانت قياساتها 10, 12, 10, 9, 14, 15, 14, 11, 13, 13. يفرض أنها تتبع التوزيع الطبيعى أوجد 99% فترة ثقة لمتوسط مجتمع تلك البيانات ؟ أوجد حجم العينة اللازم سحبها من المجتمع لتقدير المتوسط للمجتمع بحيث لا يتعدى الخطأ فى هذا التقدير (1.2) إذا كان الانحراف المعياري للمجتمع عند $\alpha = 0.05$.

السؤال الثاني (20 درجة)

البيانات الآتية تمثل جدول الاقتران لاعداد المشاهدات الخاصة بفصائل الدم و نوع الشخص (سليم - مصاب بمرض معين)

المجموع	نوع الشخص	
	مصاب	سليم
فصيلة الدم		
O	2892	698
A	2625	472
B	570	102
المجموع		

(2-1) (7-1)
1 x 2 = 2

5-991

والمطلوب اختبار ما إذا كانت هناك علاقة بين فصيلة الدم و الاصابه بهذا المرض بمستوى مغنويه 0.05.

السؤال الثالث (20 درجة)

أجريت دراسة لبحث فاعلية التنويم المغناطيسي في تخفيف الألم. الجدول الآتى يعرض النتائج لعدد من الاشخاص المختارة عشوائيا. تشير الدرجة المنخفضة إلى ألم أقل. هل القياسات الحسية أقل بعد التنويم المغناطيسي؟ اختبر عند مستوى مغنويه 5%.

قبل	6.6	6.5	9.0	10.3	11.3	8.1	6.3	11.6
بعد	6.8	2.4	7.4	8.5	8.1	6.1	3.4	2.0

ثم احسب 90% فترة ثقة للفرق بين متوسطي مستوى الألم قبل وبعد التنويم ؟

السؤال الرابع (20 درجة)

(أ) في تجربة لاختبار كفاءة أسلوب جديد لتصنيع أحد المنتجات، تم اختيار عينة حجمها 2000 وحدة مصنعة بالأسلوب الجديد فوجد أن من بينها 80 وحدة تالفة، بينما تم اختيار 1500 وحدة مصنعة بالأسلوب القديم فوجد من بينها 75 وحدة تالفة، احسب 90% فترة ثقة للفرق بين نسبتي الوحدات التالفة المصنعة بالأسلوب القديم والأسلوب الجديد.

(ب) اختبر الزعم القائل ان نسبة التالف من الوحدات المصنعة بالاسلوب الجديد لن تقل عن 0.03 عند مستوى مغنويه 0.05.

مع العلم أن :- $F_{0.05,9} = 3.48$ و $t_{0.025,10} = 2.228$ و $Z_{0.025} = 1.96$ و $Z_{0.005} = 2.58$ و $\chi^2_{0.05,5} = 11.07$ و $F_{0.05,9,5} = 4.77$ و $t_{0.05,9} = 2.262$ و $\chi^2_{0.014} = 14.8602$ و $\chi^2_{0.025,8} = 17.535$ و $Z_{0.05} = 1.654$ و $t_{0.975,9} = 2.262$ و $t_{0.05,7} = 1.8946$ و $\chi^2_{0.05,3} = 7.8147$ و $t_{0.05,7} = 1.8946$

0.06

$\bar{x} - n p_0$
 $n p_0 (p_0 - 1)$